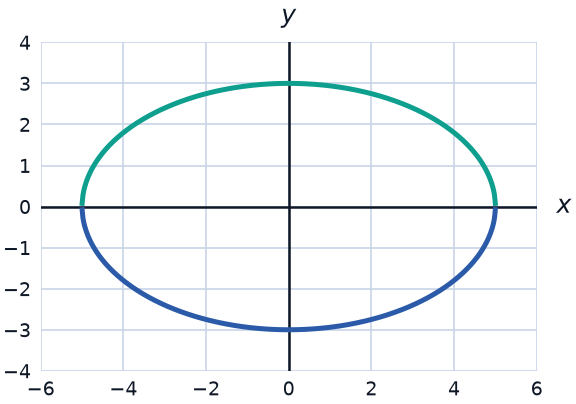
Ellipses

ClipSAT Printable Practice Suite (clipsat.org) • Topic 4.02

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Standard form

For the ellipse $\frac{x^2}{25} + \frac{y^2}{9} = 1$, find:



a the vertices

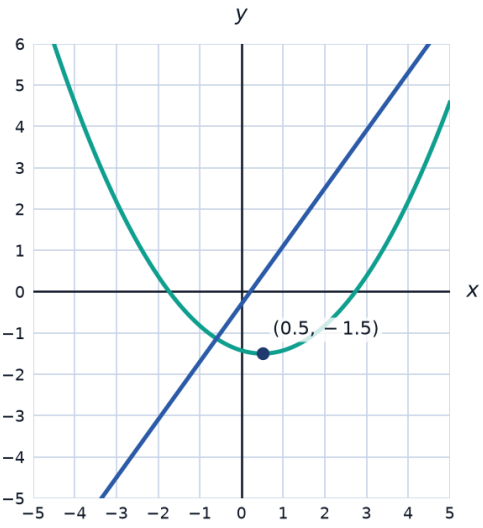
b the foci

c the endpoints of the minor axis

d the eccentricity

2 [Foundational] • Target: Standard form

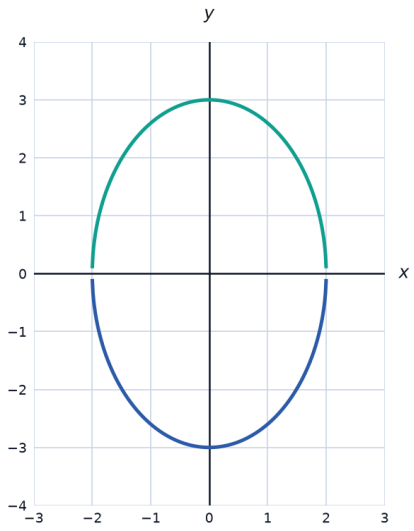
Write the standard-form equation of the ellipse with foci $(0, \pm 3)$ and vertices $(0, \pm 5)$.



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3 [Foundational] • Target: Standard form

Consider the ellipse $9x^2 + 4y^2 = 36$.



a Write the equation in standard form.

b State the lengths of the major and minor axes.

c Find the coordinates of the foci.

4 [Foundational] • Target: Applications

A whispering gallery has an elliptical ceiling 40 m long and 24 m high at the center (so $a = 20$ and $b = 12$, with the major axis horizontal). Two people stand at the foci. How far apart are they, to one decimal place?

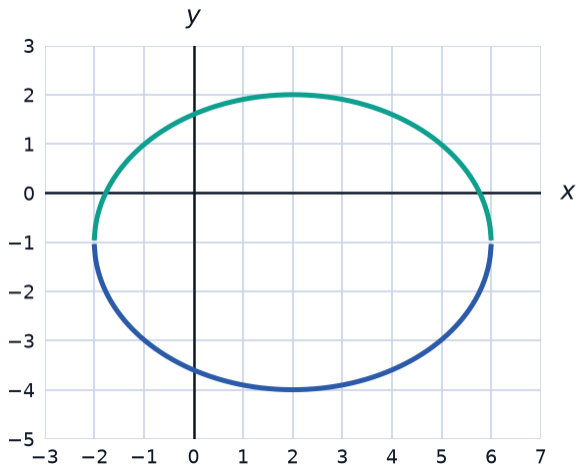
5 [Foundational] • Target: Applications

An ellipse centered at the origin has a horizontal major axis of length 12 and passes through the point $(3, 2\sqrt{3})$. Find its equation.

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6 [Foundational] • Target: Ellipses centered away from the origin

In an analytical model, for the ellipse $\frac{(x-2)^2}{16} + \frac{(y+1)^2}{9} = 1$, find:



a the center **b** the vertices **c** the foci

7 [Foundational] • Target: Ellipses centered away from the origin

Complete the square to write $4x^2 + y^2 - 8x + 6y + 9 = 0$ in standard form, and state the center.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Eccentricity and orbital applications

Compute the eccentricity of each ellipse, and state which is more circular (closer to $e = 0$):

a $\frac{x^2}{100} + \frac{y^2}{36} = 1$ **b** $\frac{x^2}{25} + \frac{y^2}{24} = 1$

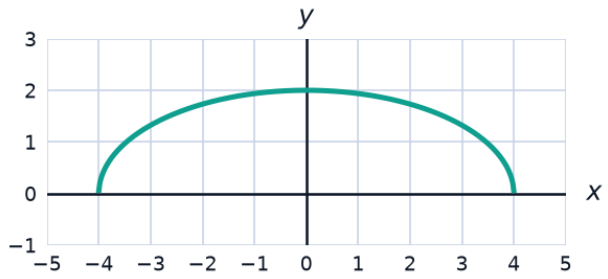
9 [Intermediate] • Target: Eccentricity and orbital applications

A planet's elliptical orbit has semi-major axis $a = 150$ million km and eccentricity $e = 0.02$. Find the distance from the center to a focus, c, and hence the closest approach distance to the sun ($a - c$), to two decimal places (in millions of km).

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10 **[Intermediate]** • **Target: Graphing an ellipse**

The upper half of the ellipse $\frac{x^2}{16}+\frac{y^2}{4}=1$ is shown (as $y=2\sqrt{1-x^2/16}$). Use it to estimate the y-value at $x=2$, then verify algebraically.

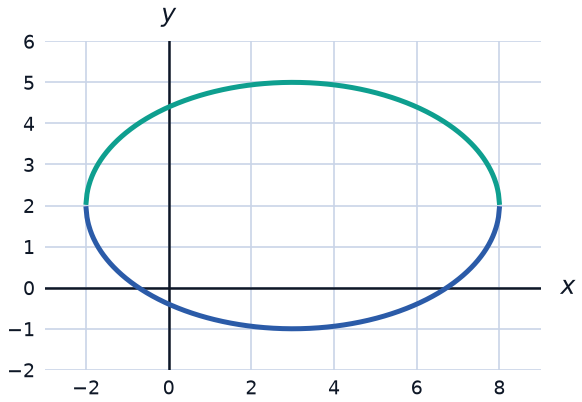


11 **[Intermediate]** • **Target: The ellipse's defining property**

State the geometric (sum-of-distances) definition of an ellipse, and use it to verify that the point $(0,3)$ lies on the ellipse $\frac{x^2}{16}+\frac{y^2}{9}=1$ with foci $(\pm\sqrt{7},0)$.

12 **[Intermediate]** • **Target: Writing equations from a description**

Write the standard-form equation of an ellipse centered at $(3,2)$ with a horizontal major axis of length 10 and minor axis of length 6.



13 **[Intermediate]** • **Target: Writing equations from a description**

Write the standard-form equation of an ellipse with vertical major axis, center $(-1,4)$, $a=7$, and $b=2$.

14 **[Intermediate]** • **Target: Systems with ellipses**

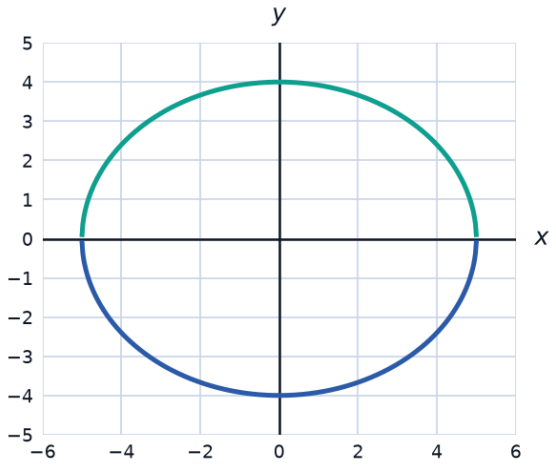
Find the points of intersection of $\frac{x^2}{9}+\frac{y^2}{4}=1$ and the line $y=x$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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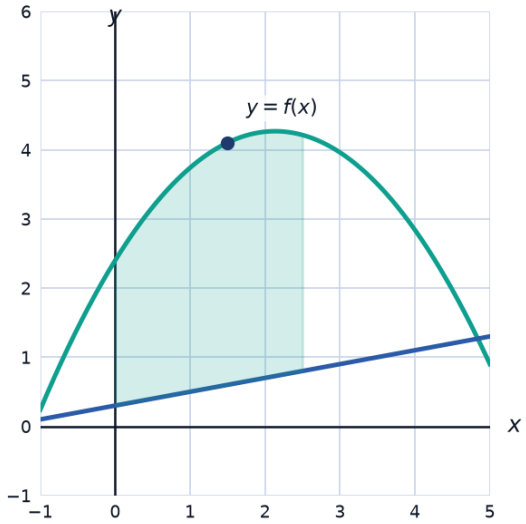
15 [Challenging] • Target: Area of an ellipse • Cross-Lesson: 9.02 Geometric Sequences and Series

State the formula for the area of an ellipse with semi-axes a and b, and use it to find the area of $\frac{x^2}{25} + \frac{y^2}{16} = 1$.



16 [Challenging] • Target: Area of an ellipse • Cross-Lesson: 9.03 The Binomial Theorem

An elliptical track for a running field has a semi-major axis of 60 m and semi-minor axis of 40 m. Find the exact area enclosed by the track, and give a decimal approximation to the nearest square metre.



17 [Challenging] • Target: Comparing ellipses of the same eccentricity • Cross-Lesson: 1.01 Functions and Function Notation

Two ellipses both have eccentricity $e = 0.6$: one has $a = 10$, the other has $a = 20$. Find b for each, and explain why they are the same shape but different sizes (similar ellipses).

18 [Challenging] • Target: Systems with ellipses • Cross-Lesson: 1.02 Domain and Range

Find the standard equation of the ellipse centered at the origin with horizontal semi-major axis $a = 5$ and semi-minor axis $b = 3$.

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19 [Challenging] • Target: Area of an ellipse • Cross-Lesson: 1.03 Average Rate of Change

Find the 6th term in the expansion of $(3x - 2y^2)^8$ in fully simplified form.

20 [Challenging] • Target: Comparing ellipses of the same eccentricity • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Decompose the vector $\mathbf{u} = \langle 6, 2 \rangle$ into components parallel and orthogonal to $\mathbf{v} = \langle 3, 4 \rangle$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Standard form

The standard equation of an ellipse centered at origin with semi-axes $a = 4, b = 3$ along x, y respectively is:

a. $\frac{x^2}{4} + \frac{y^2}{3} = 1$

b. $x^2 + y^2 = 7$

c. $\frac{x^2}{16} + \frac{y^2}{9} = 1$

d. $\frac{x^2}{9} + \frac{y^2}{16} = 1$

22 [Foundational] • Target: Applications

An ellipse $\mathcal{E}$ and a hyperbola $\mathcal{H}$ are confocal, sharing the foci $F_1(-4, 0)$ and $F_2(4, 0)$. The eccentricity of $\mathcal{E}$ is $\frac{4}{5}$, and the eccentricity of $\mathcal{H}$ is 2. If P is an intersection point of $\mathcal{E}$ and $\mathcal{H}$ in the first quadrant, what is the area of ΔPF_1F_2?

a. $6\sqrt{3}$

b. $12\sqrt{3}$

c. $3\sqrt{3}$

d. $4\sqrt{5}$

23 [Foundational] • Target: Ellipses centered away from the origin

For the ellipse $b^2x^2 + a^2y^2 = a^2b^2$, confirm this equals standard form.

a. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

b. $x^2 + y^2 = a^2 + b^2$

c. $ax + by = ab$

d. $b^2 + a^2 = 1$

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24 [Foundational] • Target: Eccentricity and orbital applications

The standard form of a circle centered at (h, k) with radius r is:

a. $x^2/r + y^2/r = 1$

b. $(x + h)^2 + (y + k)^2 = r^2$

c. $(x - h)^2 + (y - k)^2 = r^2$

d. $x^2 + y^2 = r$

25 [Foundational] • Target: Graphing an ellipse

Write the standard form of the parabola with vertex at origin opening upward with $p = 3$.

a. $x^2 = 12y$

b. $y^2 = 3x$

c. $x^2 = 3y$

d. $y^2 = 12x$

26 [Foundational] • Target: The ellipse's defining property

Find the polar form of the conic with focus at origin, directrix $x = d > 0$, eccentricity e.

a. $r = \frac{d}{1 - e\cos\theta}$

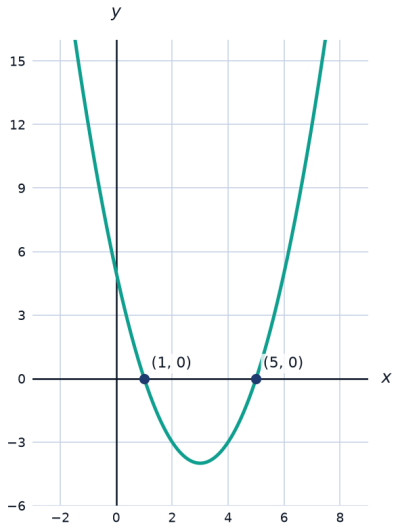
b. $r = \frac{e}{1 + d\cos\theta}$

c. $r = \frac{d}{e\cos\theta}$

d. $r = \frac{ed}{1 + e\cos\theta}$

27 [Foundational] • Target: Writing equations from a description

The graph of the quadratic function $f(x) = x^2 - 6x + 5$ is shown in the standard (x, y) coordinate plane below. A horizontal line $y = k$ intersects the graph of $f(x)$ at two points that are 8 units apart. What is the value of k?



a. 12

b. 8

c. 4

d. -3

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28 [Foundational] • Target: Systems with ellipses

Let m be a real parameter. Consider the quadratic equation $z^2 - 2mz + (m^2 + 3m - 6) = 0$ with complex roots z_1 and z_2. In the complex plane, the origin $O(0)$ and the points representing z_1 and z_2 form the vertices of a non-degenerate equilateral triangle. What is the sum of the areas of all possible equilateral triangles that can be formed in this way?

a. $15\sqrt{3}$

b. $12\sqrt{3}$

c. $\frac{15\sqrt{3}}{2}$

d. $9\sqrt{3}$

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Area of an ellipse

A wire of length 4 m is bent to form a rectangle with adjacent sides of lengths X m and $(2 - X)$ m, where X is a continuous random variable with probability density function $f(x) = \begin{cases} \frac{3}{8}x^2, & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$ Given that the area of the rectangle is at least $\frac{3}{4}$ m^2, find the probability that the length of the shorter side is at most $\frac{2}{3}$ m.

a. $\frac{127}{351}$

b. $\frac{74}{351}$

c. $\frac{37}{351}$

d. $\frac{127}{640}$

30 [Intermediate] • Target: Comparing ellipses of the same eccentricity

In the standard (x, y) coordinate plane, a triangle has vertices at $A(0, 0)$, $B(8, 0)$, and $C(2, 6)$. A vertical line with equation $x = k$, where $0 < k < 8$, divides the triangle into two regions of equal area. What is the value of k?

a. $8 - 2\sqrt{6}$

b. 4

c. $\frac{10}{3}$

d. $2\sqrt{6}$

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31 [Intermediate] • Target: Standard form

The length, X mm, of components manufactured by an industrial process is normally distributed with mean μ mm and standard deviation σ mm. A quality assurance engineer tests the null hypothesis $H_0 : \mu = 20.0$ against the alternative hypothesis $H_1 : \mu > 20.0$ at the 5% significance level using an initial random sample of size $n = 25$. The critical region for the sample mean $\bar{X}$ is found to be $\bar{X} \geq 20.658$. Using the standard normal critical value $z_{0.05} = 1.645$: 1. Find the value of σ. 2. Hence, determine the minimum sample size, m, required so that in a subsequent test of $H_0 : \mu = 20.0$ against $H_1 : \mu > 20.0$ at the same 5% significance level, the probability of committing a Type II error when the true mean is $\mu = 21.0$ is at most 0.05.

a. 44

b. 43

c. 11

d. 22

32 [Intermediate] • Target: Applications

In the standard (x, y) coordinate plane, the vertices of ΔOAB are $O(0, 0)$, $A(6, 0)$, and $B(2, 4)$. A vertical line with equation $x = k$, where $0 < k < 6$, divides ΔOAB into two regions of equal area. What is the value of k?

a. $6 - 2\sqrt{3}$

b. $\sqrt{6}$

c. $\frac{8}{3}$

d. 3

33 [Intermediate] • Target: Ellipses centered away from the origin

Consider the system of linear equations below: $\{3x - 2y = 8; kx + 6y = 5$ For which of the following values of the constant k does the system have no solution (x, y)?

a. -9

b. -4

c. 4

d. 9

34 [Intermediate] • Target: Eccentricity and orbital applications

If the real numbers x and y satisfy the system of equations $x^3 - 3xy^2, = 10; 3x^2y - y^3, = 5$ what is the value of $x^2 + y^2$?

a. $\sqrt{5}$

b. 5

c. $5\sqrt{5}$

d. 25

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35 [Intermediate] • Target: Graphing an ellipse

A particle's velocity v m/s at time t seconds is given by $v = \frac{1}{2}t^2 - 3t + 5$. The displacement of the particle from a fixed origin is s metres. It is known that at $t = 2$, the displacement $s = 7$. The displacement function $s(t)$ can be expressed in the form $s(t) = At^3 + Bt^2 + Ct + D$. Find the value of A and the value of D.

a. A = 1/6, D = 5/3

b. A = 1/6, D = 10/3

c. A = 1/3, D = 5/3

d. A = 1/3, D = 10/3

36 [Intermediate] • Target: The ellipse's defining property

In the system of linear equations below, k is a real constant. $(k-1)x + 6y, = 18; 2x + ky, = 12$ For which value of k does the system have infinitely many solutions (x, y)?

a. -4

b. -3

c. 3

d. 4

37 [Intermediate] • Target: Writing equations from a description

In the standard (x, y) coordinate plane, at what values of x do the graphs of $y = x^2 - 3x$ and $y = 2x - 4$ intersect?

a. $x = 1$ and $x = 4$

b. $x = -4$ and $x = -1$

c. $x = -2$ and $x = 2$

d. $x = 2$ and $x = 3$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Systems with ellipses • Cross-Lesson: 1.02 Domain and Range

In the standard (x, y) coordinate plane, what is the length of the major axis of the ellipse defined by the equation $\frac{(x-4)^2}{49} + \frac{(y+2)^2}{16} = 1$?

a. 7

b. 8

c. 14

d. 49

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39 [Challenging] • Target: Area of an ellipse • Cross-Lesson: 1.03 Average Rate of Change

A circle has its center at the point $(2, -3)$ and a radius of 5 units. Which of the following is the standard form equation of this circle?

a. $(x-2)^2+(y+3)^2=25$

b. $(x+2)^2+(y-3)^2=5$

c. $(x-2)^2+(y+3)^2=5$

d. $(x+2)^2+(y-3)^2=25$

40 [Challenging] • Target: Comparing ellipses of the same eccentricity • Cross-Lesson: 2.01 Right-Triangle Trigonometry

Derive the reflection property: light from one focus of an ellipse reflects to the other.

a. By Snell's law variant: angle of incidence = angle of reflection using the definition $PF_1+PF_2=2a$ and tangent bisects angle F_1PF_2

b. Only for circles

c. Only for parabolas

d. Arbitrary property

41 [Challenging] • Target: Standard form • Cross-Lesson: 2.02 Radian Measure and Arc Length

Find the polar equation of an ellipse with focus at the origin, semi-major $a=5$, $e=0.6$.

a. $r=\frac{5}{1-\cos\theta}$

b. $r=\frac{5}{1+0.6\cos\theta}$

c. $r=\frac{3.2}{1-0.6\cos\theta}$

d. $r=\frac{8}{1+0.6\cos\theta}$

42 [Challenging] • Target: Applications • Cross-Lesson: 2.03 Graphs of Sine and Cosine Functions

Convert the hyperbola $4x^2-9y^2=36$ to standard form.

a. $\frac{x^2}{9}-\frac{y^2}{4}=1$

b. $x^2-y^2=9$

c. $\frac{x^2}{36}-\frac{y^2}{36}=1$

d. $\frac{x^2}{4}-\frac{y^2}{9}=1$

43 [Challenging] • Target: Ellipses centered away from the origin • Cross-Lesson: 2.04 Law of Sines and Law of Cosines

Convert to standard form: $(3+4i)(1-2i)$.

a. $3-8i$

b. $11-2i$

c. $11+2i$

d. $7+2i$

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44 [Challenging] • Target: Eccentricity and orbital applications • Cross-Lesson: 3.01 The Unit Circle and Exact Values

Find the parametric equations for the ellipse $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

a. $x = 4\sin t, y = 3\cos t$

b. $x = 4t, y = 3t$

c. $x = 4\cos t, y = 3\sin t$

d. $x = \cos t, y = \sin t$

45 [Challenging] • Target: Graphing an ellipse • Cross-Lesson: 3.02 Fundamental Trigonometric Identities

A system of 3 equations in 3 unknowns is represented by $AX = B$ where A is:

a. 3×3

b. $3 \times B$

c. 3×1

d. 1×3

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